THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · EASY LEVEL CHAPTER TEST SET 1

Chapter 6: **How Forces Affect Motion** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Force is a: 1

(a) speed **(b)** mass **(c)** type of energy **(d)** push or pull

2. A heavier body has: 1

(a) same inertia as a lighter one **(b)** no inertia

(c) more inertia **(d)** less inertia

3. A force of 10 N on a mass of 5 kg gives acceleration: 1

(a) 0.5 m/s squared **(b)** 15 m/s squared **(c)** 50 m/s squared **(d)** 2 m/s squared

4. The SI unit of force: 1

(a) metre **(b)** joule **(c)** newton **(d)** kg

5. A rocket works on the: 1

(a) law of gravitation only **(b)** third law

(c) second law **(d)** first law

6. A mass of 2 kg accelerates at 3 m/s squared. Force: 1

(a) 5 N **(b)** 1.5 N **(c)** 6 N **(d)** 0.67 N

7. The first law is also called the law of: 1

(a) gravitation **(b)** energy **(c)** inertia **(d)** momentum

8. A body at rest remains at rest due to: 1

(a) inertia **(b)** friction **(c)** force **(d)** gravity

9. A mass of 10 kg has a weight (g = 10): 1

(a) 10 N **(b)** 1 N **(c)** 100 N **(d)** 1000 N

10. Inertia depends on: 1

(a) volume **(b)** colour **(c)** shape **(d)** mass

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. State Newton's first law. 2

12. What is momentum? 2

13. Differentiate between mass and weight in two points. 2

14. State Newton's second law. 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. Give three examples of inertia of rest. 3

16. Find the force on a 1000 kg car with an acceleration of 3 m/s squared. 3

17. Find the net force when 10 N east and 4 N west act on a body. 3

OR

Give three examples of inertia of motion. 3

18. Find the momentum of a 5 kg body with a velocity of 4 m/s. 3

OR

Find the net force when 5 N and 7 N act in the same direction. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. State Newton's laws of motion with an example for each. 5

OR

Explain the working of a rocket. 5

20. Explain inertia with three examples. 5

OR

Explain how seat belts and airbags help passengers. 5

END OF QUESTION PAPER Total: 40 Marks

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